

Identifying Risk-Neutral Densities from an Empirical Measure of Listed Quotes

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Abstract

European option prices of a single maturity encode the risk-neutral density of the underlying at expiry. Differentiating those prices twice is ill-posed on a finite strike list. This paper rescales the calls to a complementary distribution function and smooths the jump measure of that function with a Gaussian kernel. The density estimator is minus the forward times the derivative of the resulting kernel density, one term per strike. The kernel cdf of the same measure is a call interpolant. Completing the unquoted tails, placing masses at cell centers, and combining two kernel widths are quadrature corrections. Inside a filling strike grid the estimator is uniformly consistent for exact quotes. Over the whole line the signed estimator integrates to zero and its first moment equals the forward. On exact Heston quotes the corrections bring the integrated squared error to 5.0×10^{-9} . When the quotes contain butterfly arbitrage the calls are replaced by their convex projection, the wings are held at the wing implied volatility, and the density is a mixture of the twiced Gaussian and a wider sinc kernel. The bandwidth factor, the mixture weight, and the sinc multiple are chosen by pricing cross-validation after that pipeline is assembled. On thirty noisy Heston replications the mean integrated squared error is 6.6×10^{-4} . The median is 4.0×10^{-4} . Under variance gamma the mean is 4.2×10^{-3} . An equal mixture with the sinc fixed at three times the density-scale bandwidth, chosen instead by integrated squared error on a pilot of eight replications, reaches 2.8×10^{-4} and 1.3×10^{-3} . Pricing cross-validation does not find those constants. On listed slices the unsmoothed projection remains closer to the quotes. The kernel step is what returns a density.

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1. Introduction

Breeden and Litzenberger (1978) identified the risk-neutral density of the underlying at a fixed expiry with the second strike derivative of the corresponding European call, scaled by a discount factor. The identity assumes a twice differentiable price curve. A market supplies a finite ordered list of strikes and a quote at each knot. Numerical second differentiation of that list is ill-posed: interpolants oscillate, change sign, and depend on the interpolating scheme. The literature responds with parametric mixtures and Gram–Charlier expansions, implied-volatility interpolants, local polynomials of the call, and Bondarenko (2003)’s positive convolution approximation, which forces the density to be a mixture of a positive kernel with a discrete probability fitted to prices. Section 2 reviews these approaches.

This paper estimates the density from a single-maturity strip by rescaling the calls to a complementary distribution function, replacing the empirical jump measure by a Gaussian kernel, and inverting the associated characteristic function. For a Gaussian the inversion is a finite sum, one term per strike. Completing the unquoted tails, placing each mass at the center of its cell, and combining two kernel widths are quadrature corrections. As the strike mesh fills a compact interval and the bandwidth shrinks more slowly than the mesh, the estimator converges to the true density on the interior of that interval.

The usual split in this literature is that interpolants of the call or of implied volatility reprice listed quotes well and do not yield a usable density, while methods that are densities by construction cannot interpolate a dense, slightly arbitrageable chain. Bondarenko’s method is a sieve: with a kernel width that does not shrink as strikes are added, it does not become a local smoother, and it is a density by construction. Yatchew and Härdle (2006) replace the quoted calls by a decreasing convex list; at a zero roughness penalty the second differences of that list are a discrete nonnegative measure, not a curve. Aït-Sahalia and Duarte (2003) smooth the same constrained list with a local polynomial, so the density is nonnegative by design and the call cannot match quotes that violate convexity. The estimator of this paper returns a closed-form estimate of f_Q and, from the same kernel, a call interpolant. They are different objects. Over the whole line

the signed density integrates to zero and its first moment equals the forward; the prices are read from the kernel cdf. Sections 7 and 8 compare both objects with the benchmarks above, with tuning parameters chosen by cross-validation on pricing error, and with a raw SVI slice (Gatheral, 2006; Gatheral and Jacquier, 2014). On exact quotes the quadrature corrections are effective. Under noise the kernel is applied only after the calls have been projected and the wings filled at constant implied volatility. A Gaussian–sinc mixture on those masses can undercut the other density estimators, if the weight and the sinc width are chosen by integrated squared error on a pilot. Pricing cross-validation, run after the same pipeline is assembled, does not recover that choice: the mean integrated squared error rises, and under variance gamma it is no longer the smallest in the comparison. On the listed slices the unsmoothed projection remains the closest call. It does not return a density curve.

The remainder of this manuscript is structured as follows. Section 2 reviews related work. Section 3 derives the characteristic function of the normalized call and the closed-form estimator, and compares it to kernel smoothing of the call. Section 4 records the quadrature corrections and the integral identities. Section 5 chooses the bandwidth. Section 6 proves in-fill consistency and illustrates it on exact Heston and variance-gamma quotes. Section 7 repeats the comparison on irregular strikes with noise in implied volatility, against retuned benchmarks and an SVI slice. Section 8 is an illustration on four listed chains from three snapshots. Section 9 concludes.

2. Related Work

The identity of Section 3 is the continuous-strike version of a spanning argument of Ross (1976): if one could trade a put at every strike, the second derivative of the put curve would be a density, and one could synthesize any sufficiently regular payoff. Dupire (1994) obtained the analogous result in calendar time, which produces a local-volatility equation. Observed markets supply neither a continuum of strikes nor noiseless quotes.

A first family of methods specifies the density of S_T as a parametric curve and fits the implied option prices to the quotes. Ritchey (1990) and Melick and Thomas (1997) take the density to be a mixture of lognormals, so that the call price is a mixture of Black–Scholes prices; Bahra (1997) and Söderlind and Svensson (1997) use related specifications for interest-rate and foreign-exchange options. Two or three components typically suffice to generate a unimodal, skewed density, but the fit is poorly identified once more than two components are admit-

ted, and the approximation to a Heston or jump-diffusion density need not vanish as more quotes arrive. Expansion methods perturb a baseline density—usually lognormal—by Gram–Charlier or Hermite terms (Jarrow and Rudd, 1982; Corrado and Su, 1996; Rompolis and Tzavalis, 2007; Xiu, 2014). The expansions can match a finite number of moments, but they are not guaranteed to stay nonnegative, and the truncated series can oscillate in the tails.

A second, and in applied work the most common, family interpolates the Black–Scholes implied-volatility function of strike and converts the interpolant back to prices before differentiating twice. Shimko (1993) fitted a quadratic in strike; Malz (1997) used a cubic spline in delta; Bliss and Panigirtzoglou (2002) a smoothing spline. Figlewski (2010) completes the tails with a generalized extreme-value distribution. Because implied volatility is a much flatter function of strike than the call itself, a low-order interpolant often has small bias when quotes are accurate. The drawback is that implied volatility is a nonlinear transformation that inflates price noise where the derivative of price with respect to implied volatility is small, i.e. in the tails. An interpolating spline then treats those noisy tail quotes as exact, and the second derivative of the resulting call curve oscillates and can become negative. Shape-constrained smoothers of the implied-volatility surface (Fengler, 2005, 2009; Gatheral and Jacquier, 2014; Ludwig, 2015) address the sign problem at the cost of a more involved optimization.

A third family estimates the call as a function of strike by kernel regression or local polynomials and obtains the second derivative from the fit. Priestley and Chao (1972) fitted a kernel to observed function values at the design points; the risk-neutral density is obtained by differentiating that fit twice. Aït-Sahalia and Lo (1998) estimated the call as a function of (S, K, T, r, q) by kernel regression; Aït-Sahalia and Lo (2000) used the estimator to recover an implied risk-aversion function. The method is data intensive and, in the original application, pooled option prices across many trading dates. Aït-Sahalia and Duarte (2003) developed a local polynomial estimator that imposes monotonicity and convexity of C in K , hence nonnegativity of the density. Related constrained least-squares estimators are studied by Yatchew and Härdle (2006) and Birke and Pilz (2009); Grith et al. (2012) survey the kernel literature. Fan and Mancini (2009) combine a parametric guide with a nonparametric correction. In each case the density is the second strike derivative of a smoother of C . The estimator of Section 3 rescales the calls to a complementary cumulative distribution function (cdf) and applies the kernel to the empirical jump measure, so the formula is a first derivative of a Gaussian kernel density and is a finite sum. Section 3.2 records the compari-

son. Sections 7 and 8 implement the cubic Priestley and Chao estimator of that section and the two-step procedure of Aït-Sahalia and Duarte: a least-squares projection of the quoted calls onto the decreasing convex cone, followed by a local polynomial of the constrained values. The pricing bandwidth of each is chosen by even/odd cross-validation. The local-cubic bandwidth keeps the second-derivative rate $n^{-1/9}$.

Bondarenko (2003) proposed the positive convolution approximation (PCA). The density is required to lie in the set of functions obtained by convolving a discrete probability with a positive kernel, and the discrete probability is chosen to minimize the sum of squared pricing errors. The kernel enforces smoothness and, together with positive weights, guarantees that the result is a density. In Bondarenko’s comparison, the method dominated mixtures of lognormals, Hermite expansions, and several regularization methods. Subsequent simulation evidence on affine jump-diffusions is consistent with this ranking (Lu, 2019). Sections 7 and 8 implement a log-price analog: forty-one lognormal components with a common width, centered on a uniform grid in log strike from the first quoted strike to the last, fitted by nonnegative least squares. Sum-to-one and forward-mean penalties are left as penalties. The weights are not rescaled after the fit, and the width is chosen by even/odd cross-validation, with the grid starting at the center spacing. The same sections choose the roughness penalty of Yatchew and Härdle by the same cross-validation. Zero is on the grid, so the interpolating projection remains available.

Rubinstein (1994) and Jackwerth and Rubinstein (1996) recovered probabilities on a binomial tree by minimizing a distance to a prior subject to pricing constraints. Stutzer (1996) and Buchen and Kelly (1996) maximized entropy. These methods produce a valid density by construction, but the finite-support discretization and the choice of prior affect the tails. Viewing the problem as a linear inverse problem, Cont and Tankov (2004) calibrated jump-diffusion models under a relative-entropy penalty; Monnier (2014) used the singular value decomposition of a restricted put operator.

A number of papers compare methods on market data, where the true density is unknown, so that the criterion is typically pricing error or stability over time (Cooper, 1999; Jondeau and Rockinger, 2000; Coutant et al., 2001; Anagnou et al., 2005; Bliss and Panigirtzoglou, 2002; Liu et al., 2007). Studies that generate options from a known process are less common. Bondarenko (2003) finds that his method outperforms mixtures, expansions, and several regularization methods on integrated squared error. Lai (2014) finds that kernel regression outperforms spline interpolation and neural nets on pricing error. Santos and

Guerra (2015) compare mixtures, Edgeworth expansions, hypergeometric functions, and a Bliss and Panigirtzoglou smoothing spline against Heston and other targets. Lu (2019) compares methods on affine jump-diffusions and again finds Bondarenko's method accurate. The typical ranking is that implied-volatility interpolants price well when quotes are clean, that mixtures are unstable once more than two components are admitted, and that Bondarenko's method is the most accurate among methods that do not use a parametric density of S_T , when the true density is known. Section 6 is in this vein for exact quotes: the targets are Heston and variance-gamma, and only the nested versions of the estimator in this paper are scored. Section 7 puts the same targets behind an irregular strike grid and one volatility point of noise, and scores the retuned benchmarks on integrated squared error. Section 8 scores four listed slices, where the true density is unknown, on pricing error.

3. Construction of the Estimator

3.1. Construction

Assume a risk-neutral measure $\mathbb{Q}$ under which S_T has a density $f_{\mathbb{Q}}$ on $(0, \infty)$ and $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$. Let r be the continuously compounded interest rate and q the continuous dividend yield. Write $F = S_0 e^{(r-q)T}$ for the forward. Under $\mathbb{Q}$,

$$\begin{aligned} S_0 e^{-qT} &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T] \\ &= e^{-rT} \int_0^{\infty} x f_{\mathbb{Q}}(x) dx, \end{aligned} \tag{1}$$

hence $\mathbb{E}^{\mathbb{Q}}[S_T] = F$. Write C for the corresponding call curve:

$$\begin{aligned} C(K) &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \\ &= e^{-rT} \int_K^{\infty} (x - K) f_{\mathbb{Q}}(x) dx. \end{aligned}$$

Now define the normalized option price

$$\tilde{C}(K) \equiv \frac{C(K)}{S_0 e^{-qT}} = \frac{\int_K^{\infty} (x - K) f_{\mathbb{Q}}(x) dx}{F}. \tag{2}$$

The normalized call price is a decreasing function of K : it is equal to 1 if $K = 0$ and equal to 0 if $K = \infty$. Thus the normalized call price, as a function

of K , is a complementary cdf, say $\tilde{C}(K) = 1 - P(K)$. The characteristic function (hereafter c.f.) of the law of P is the Fourier–Stieltjes transform

$$\varphi_P(s) = \int_{(0,\infty)} e^{isK} dP(K). \quad (3)$$

For a cdf on $[0, \infty)$ and $s \neq 0$, (3) is

$$\varphi_P(s) = 1 + is \int_0^\infty e^{isK} \tilde{C}(K) dK.$$

Substitute (2). Fubini yields

$$\int_0^\infty e^{isK} \tilde{C}(K) dK = \frac{1}{F} \int_0^\infty f_{\mathbb{Q}}(x) \left[\int_0^x e^{isK} (x - K) dK \right] dx.$$

The inner integral is $-x/(is) + (e^{isx} - 1)/(is)^2$. Hence, for $s \neq 0$,

$$\begin{aligned} \varphi_P(s) &= 1 + \frac{is}{F} \int_0^\infty f_{\mathbb{Q}}(x) \left[-\frac{x}{is} + \frac{e^{isx} - 1}{(is)^2} \right] dx \\ &= 1 - F^{-1} \mathbb{E}^{\mathbb{Q}}[S_T] + \frac{\varphi_{\mathbb{Q}}(s) - 1}{is F} \\ &= \frac{\varphi_{\mathbb{Q}}(s) - 1}{is F}, \end{aligned} \quad (4)$$

where $\varphi_{\mathbb{Q}}(s) = \mathbb{E}^{\mathbb{Q}}[e^{isS_T}]$ and $\mathbb{E}^{\mathbb{Q}}[S_T] = F$ by (1). Solving for $\varphi_{\mathbb{Q}}$, one obtains

$$\varphi_{\mathbb{Q}}(s) = 1 + is F \varphi_P(s). \quad (5)$$

Read from the right, (4) is a pricing formula: a model that supplies $\varphi_{\mathbb{Q}}$ yields φ_P , hence the complementary cdf $\tilde{C} = 1 - P$ and the call (Lewis, 2000). This paper inverts the identity: φ_P is constructed from the quotes, and $\varphi_{\mathbb{Q}}$ is recovered.

The c.f. $\varphi_{\mathbb{Q}}$ is that of S_T . Fourier inversion yields

$$f_{\mathbb{Q}}(K) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-isK} \varphi_{\mathbb{Q}}(s) ds. \quad (6)$$

Substitute (5) into (6):

$$f_{\mathbb{Q}}(K) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-isK} ds + \frac{F}{2\pi} \int_{-\infty}^\infty e^{-isK} (is) \varphi_P(s) ds.$$

The first integral is a Dirac mass at $K = 0$. For $K > 0$,

$$f_{\mathbb{Q}}(K) = \frac{F}{2\pi} \int_{-\infty}^{\infty} e^{-isK} (is) \varphi_P(s) ds. \quad (7)$$

Observed calls at strikes $0 < K_1 < \dots < K_m$ supply an empirical complementary cdf $\tilde{C}_i = C_i/(S_0 e^{-qT})$ and hence an empirical cdf $P_i = 1 - \tilde{C}_i$. Write

$$\nu_m = \sum_{i=1}^m \Delta P_i \delta_{K_i}, \quad \Delta P_i = P_i - P_{i-1}, \quad P_0 := 0. \quad (8)$$

The càdlàg interpolant of these knots is a step function P_m . Its Fourier–Stieltjes transform is the finite sum $\int e^{isK} dP_m(K) = \sum_i \Delta P_i e^{isK_i}$, which is also the c.f. of ν_m . Replace φ_P by the product of that sum with the Fourier transform of a Gaussian kernel of width h ,

$$\hat{\varphi}_P(s) = \left(\sum_{i=1}^m \Delta P_i e^{isK_i} \right) e^{-h^2 s^2 / 2}. \quad (9)$$

Formally, the estimator is (7) with φ_P replaced by $\hat{\varphi}_P$:

$$\hat{f}_{\mathbb{Q}}(K) = \frac{F}{2\pi} \int_{-\infty}^{\infty} e^{-isK} (is) \left(\sum_{i=1}^m \Delta P_i e^{isK_i} \right) e^{-h^2 s^2 / 2} ds.$$

The integrand is a finite sum. The Fourier transform of $e^{isK_i} e^{-h^2 s^2 / 2}$ is the $N(K_i, h^2)$ density

$$\kappa_h(u) = \frac{1}{h\sqrt{2\pi}} \exp\left(-\frac{u^2}{2h^2}\right), \quad (10)$$

and multiplication by is is differentiation up to sign: if $\varphi(s) = \int e^{isu} \kappa_h(u - K_i) du$, then the transform of $\kappa'_h(\cdot - K_i)$ is $-is \varphi(s)$, so the transform of $is \varphi(s)$ is $-\kappa'_h(\cdot - K_i)$. Therefore

$$\hat{f}_{\mathbb{Q}}(K) = -F \sum_{i=1}^m \Delta P_i \kappa'_h(K - K_i) = F \sum_{i=1}^m \Delta P_i \frac{K - K_i}{h^2} \kappa_h(K - K_i), \quad (11)$$

where $\kappa'_h(u) = -(u/h^2) \kappa_h(u)$. Equation (11) is a closed-form sum. It is $-F$ times the first derivative of the Gaussian kernel density of ν_m . The kernel density itself is a Gaussian mixture. The estimator of $f_{\mathbb{Q}}$ is not: each term is a Gaussian density multiplied by $(K - K_i)/h^2$. The interpolant of the complementary-call cdf is the

kernel cdf of the same measure, and that interpolant does not differentiate the kernel. Let $\mathcal{K}$ denote the standard Gaussian cdf. Then

$$\hat{P}(K) = \sum_{i=1}^m \Delta P_i \mathcal{K}((K - K_i)/h), \quad (12)$$

and calls are $\hat{C} = S_0 e^{-qT} (1 - \hat{P})$. Differentiating $\hat{P}$ twice and multiplying by $-F$ recovers (11).

3.2. Comparison to Naive Kernel Smoothing

In the continuum, Breeden and Litzenberger plus a kernel is unambiguous. If C is the true call curve,

$$(C * \kappa_h)'' = C'' * \kappa_h = e^{-rT} (f_{\mathbb{Q}} * \kappa_h). \quad (13)$$

Smoothing C and differentiating twice is a kernel smooth of $f_{\mathbb{Q}}$. On a strike list the discretization matters.

A Priestley and Chao estimator treats the observations as levels C_i and weights them by the design spacing $\Delta K_i = K_i - K_{i-1}$ ($K_0 := K_1 - (K_2 - K_1)$):

$$\hat{C}^{\text{PC}}(K) = \sum_{i=1}^m \Delta K_i C_i \kappa_h(K - K_i), \quad \hat{f}_{\mathbb{Q}}^{\text{PC}}(K) = e^{rT} \sum_{i=1}^m \Delta K_i C_i \kappa_h''(K - K_i), \quad (14)$$

where $\kappa_h''(u) = (u^2/h^2 - 1)h^{-2}\kappa_h(u)$. The equivalent kernel for the density is κ'' . The interpolant is a mixture of the prices.

That sum is a Riemann approximation to (13). It treats the observed levels as atoms of a measure. They are not: C is a smooth function of K , observed at the knots. Convolution requires a function on the line. A linear interpolant of C has second derivative zero almost everywhere and Dirac masses at the kinks, so it cannot recover $f_{\mathbb{Q}}$. The usual construction is a cubic spline C^{cs} through the quotes, extended by linear wings of slope in $[-e^{-rT}, 0]$:

$$\hat{f}_{\mathbb{Q}}^{\text{cs}}(K) = e^{rT} (C^{\text{cs}} * \kappa_h)''(K). \quad (15)$$

A cubic is the lowest-degree interpolant whose second derivative is not identically zero on each cell; the same degree appears in local-polynomial estimation of a second derivative (Aït-Sahalia and Duarte, 2003). Without the interpolant, (14) places a large weight on κ'' wherever C is large, including deep in the money where C is nearly linear and C'' is nearly zero.

Equation (11) does not need a spline. The object being smoothed is already a discrete measure, $\nu_m = \sum_i \Delta P_i \delta_{K_i}$. Convolution with κ_h is the finite mixture $\sum_i \Delta P_i \kappa_h(\cdot - K_i)$, and one derivative of that mixture is (11). The càdlàg interpolant of P is a step function; smoothing a step function is kernel density estimation, not kernel regression of a curve.

On an equal grid with already monotone C , $\Delta C_i \approx C'(K_i)\delta$, so (14) and (11) are close. They are not close when C is large and locally linear, as a deep in-the-money call is. Section 8 implements the cubic interpolant (15) on listed chains.

4. Three Refinements

4.1. Completing the Tails

The empirical measure (8) puts mass $P(K_1)$ at the first quoted strike and mass $1 - P(K_m)$ nowhere. Both ends are wrong as a cdf on $(0, \infty)$.

On $(0, K_1]$ a deep in-the-money call is intrinsic, $C(K) = e^{-rT}(F - K)$, hence $\tilde{C}(K) = 1 - K/F$ and $P(K) = K/F$. That is a straight line through the origin. The estimator matches the observed $P(K_1)$ rather than K_1/F ,

$$P(K) = P(K_1) \frac{K}{K_1}, \quad 0 < K \leq K_1. \quad (16)$$

When the first quote is at intrinsic, $P(K_1) = K_1/F$ and (16) is exactly K/F . The line is replaced by $n_L = \text{round}(K_1/\delta)$ equal jumps of size $P(K_1)/n_L$ on a uniform grid of $(0, K_1]$, where δ is the median quoted gap, and the original jump $\Delta P_1 = P(K_1)$ at K_1 is removed. That choice continues the quoted mesh through the splice at K_1 .

On the right, $\tilde{C}(\infty) = 0$ so $P(\infty) = 1$. If $C_m > 0$, a European call still has to fall to zero. The residual call is continued linearly with slope in $[-e^{-rT}, 0)$ until $C = 0$ at an intercept K_{end} . That wing is a line. The last listed gap is much coarser than δ , so the knots are a quadrature of the line rather than a continuation of the quoted mesh. Eight knots per last gap,

$$n_R = \text{round}(8(K_{\text{end}} - K_m)/(K_m - K_{m-1})),$$

are equally spaced in $(K_m, K_{\text{end}}]$ and converted to $P = 1 - \tilde{C}$. If $C_m = 0$, $P(K_m) = 1$ already and this wing is empty.

Only the left completion changes the estimator on the Heston and variance-gamma chains below: the first quoted call is at intrinsic, and the last quoted call is already zero (dense) or of order 10^{-5} (sparse).

4.2. Midpoint Rule

Placing each jump ΔP_i at the right endpoint of its cell $(K_{i-1}, K_i]$ is a rectangle rule for the Stieltjes integral against P . The midpoint rule puts the same mass at the cell center. Write $K_0 := 0$ and

$$\tilde{K}_j = \frac{1}{2}(K_{j-1}^* + K_j^*),$$

where $K_1^* < \dots < K_N^*$ is the completed strike list. For a Gaussian integrand on an equally spaced grid the midpoint error is smaller than any power of δ/h . Combined with both tails,

$$\hat{f}_{\mathbb{Q}}(K) = F \sum_{j=1}^N \Delta_j \frac{K - \tilde{K}_j}{h^2} \kappa_h(K - \tilde{K}_j), \quad (17)$$

with κ_h as in (10) and Δ_j the jump on $(K_{j-1}^*, K_j^*]$. Equation (11) is the same sum with $\tilde{K}_j$ replaced by the right endpoints K_j^* . The interpolant on that support is (12) with each K_i replaced by $\tilde{K}_j$,

$$\hat{P}(K) = \sum_{j=1}^N \Delta_j \mathcal{K}((K - \tilde{K}_j)/h). \quad (18)$$

4.3. Twicing

The Gaussian kernel in (17) has second moment h^2 , so the leading bias of $\hat{f}_{\mathbb{Q}}$ is $O(h^2)$. When the quotes are treated as exact there is no observation-error variance to inflate, and a higher-order kernel is admissible. Twicing (Schucany and Sommers, 1977) replaces the kernel transform $e^{-h^2 s^2/2}$ in (9) by

$$2e^{-h^2 s^2/2} - e^{-h^2 s^2}, \quad (19)$$

which is the Fourier transform of $2\kappa_h - \kappa_{h\sqrt{2}}$. The equivalent kernel has vanishing second moment, so the $O(h^2)$ bias becomes $O(h^4)$. In closed form,

$$\hat{f}_{\mathbb{Q}}^{\text{tw}}(K) = 2\hat{f}_{\mathbb{Q},h}(K) - \hat{f}_{\mathbb{Q},h\sqrt{2}}(K), \quad (20)$$

and the same combination is applied to the interpolant (18), $\hat{P}^{\text{tw}} = 2\hat{P}_h - \hat{P}_{h\sqrt{2}}$. The equivalent kernel takes negative values, so $\hat{f}_{\mathbb{Q}}^{\text{tw}}$ is not a density. It is still $-F$ times the derivative of a two-width Gaussian mixture of the jumps.

Twicing a truncated measure amplifies leakage from the missing tails. It is used only on the completed, midpoint-placed support. Equation (20), applied to (17), is the estimator used below.

4.4. Integrals

The kernel density of the completed midpoint measure, $m_h = \nu_N * \kappa_h$, integrates to $\sum_j \Delta_j$. When those jumps are the jumps of a cdf the sum equals one, and m_h is a Gaussian mixture, one component per strike. The density estimator is not the mixture. It is $\hat{f}_{\mathbb{Q}} = -F m'_h$. Integration by parts, using $K m_h(K) \rightarrow 0$ as $|K| \rightarrow \infty$, gives

$$\int_{-\infty}^{\infty} \hat{f}_{\mathbb{Q}}(K) dK = 0, \quad \int_{-\infty}^{\infty} K \hat{f}_{\mathbb{Q}}(K) dK = F \sum_j \Delta_j. \quad (21)$$

The same identities hold after twicing. The kernel $2\kappa_h - \kappa_{h\sqrt{2}}$ integrates to one, so $\hat{f}_{\mathbb{Q}}^{\text{tw}}$ is the derivative of a function that integrates to $\sum_j \Delta_j$, whether or not that function stays nonnegative. The signed estimator therefore matches the forward and misses the probability. On an exact dense Heston strip, with the mesh bandwidth and twicing, the two integrals on $[-2F, 8F]$ equal 0 and F to machine precision, and the positive and negative parts on that window each integrate to about 1.07.

Theorem 1 is uniform approximation on compact subsets of an interval (A, B) that the quotes cover. It does not put the missing unit of mass back. The negative part that cancels $\int_{(A,B)} \hat{f}_{\mathbb{Q}}$ lies outside that interval, where m_h is still rising. Call prices below are not integrals of $\hat{f}_{\mathbb{Q}}$. They are read from the kernel cdf $\hat{P}$, clipped to $[0, 1]$. Clipping $\hat{P}$ and discarding the negative part of $\hat{f}_{\mathbb{Q}}$ are different operations.

5. Choice of Bandwidth

The width h is the only tuning constant in (20). Two rules are used below.

With exact quotes the bias is the usual $O(h^2)$ kernel bias, or $O(h^4)$ after twicing, and the remaining discretization error after midpoints is smaller than any power of δ/h on an equally spaced Gaussian grid. For a Gaussian it is enough that h be a small multiple of the typical quoted gap. Write δ for the median of $\{K_i - K_{i-1}\}$ on the quoted strikes, not on the filled wings, and take

$$h = \min(1.2 \delta, 0.30 (K_m - K_1)). \quad (22)$$

The cap binds only on a very short chain. On an equally spaced grid every gap equals δ . This is the rule in Section 6.1, where the quotes are exact.

When quotes are noisy, (22) inherits the noise in the jumps. A listed chain is a noisy sample of the call curve, not an in-fill of an exact function. The object being smoothed is then closer to a density of the jumps than to a second derivative from

a vanishing mesh. Replacing the mesh by the width of S_T and using the Gaussian exponent of Silverman (1986), $n^{-1/5}$, yields

$$h = 1.06 F \hat{\sigma}_{\text{ATM}} \sqrt{T} n^{-1/5}, \quad (23)$$

where $\hat{\sigma}_{\text{ATM}}$ is the at-the-money (ATM) Black–Scholes implied volatility, the median of contracts with $|K - F| \leq 0.03F$ (or the median of all finite implied volatilities if that window is empty), and n is the number of masses after the right wing of Section 4.1, not the left filler and not necessarily the number of listed quotes. The scale is the scale of S_T , not the standard deviation of the jumps Δ_j . The ordinary Silverman rule computed from those jumps oversmooths, because that standard deviation is of the order of the chain width. This is the rule in Section 8.

Theorem 1 requires $h \rightarrow 0$ and $\delta/h^{3/2} \rightarrow 0$ on a general mesh, or $\delta/h \rightarrow 0$ on an equally spaced Gaussian grid. On a sequence of grids that fill a fixed interval with mesh $\delta \propto n^{-1}$, the density-scale rule has $h \propto n^{-1/5}$. Then $h \rightarrow 0$ and $\delta/h^{3/2} = O(n^{-7/10}) \rightarrow 0$, so (23) satisfies the theorem for exact quotes. The listed chains below are noisy, and the theorem is silent on observation error. The mesh rule keeps $h/\delta = 1.2$, so δ/h does not tend to zero and the polynomial midpoint bound $O(\delta^2/h^3)$ does not vanish. Equation (22) is not covered by Theorem 1. It is a fixed-ratio quadrature for exact, equally spaced quotes, used on the chains of Section 6.1.

6. In-Fill Consistency

Write P for the complementary-call cdf of Section 3, so that $f_{\mathbb{Q}} = -F P''$ on $(0, \infty)$. The midpoint estimator (17) is

$$\hat{f}_{\mathbb{Q}}(K) = -F(\nu_N * \kappa_h)'(K),$$

where $\nu_N = \sum_j \Delta_j \delta_{\tilde{K}_j}$ is the completed, midpoint-placed empirical measure of P .

Theorem 1. *Let $f_{\mathbb{Q}}$ be four times continuously differentiable on $(0, \infty)$ with $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$, and let C be the corresponding call curve. Fix a compact interval $[A, B] \subset (0, \infty)$. Consider a sequence of exact quote grids $0 < K_1^{(m)} < \dots < K_m^{(m)}$ with mesh $\delta_m = \max_i (K_i^{(m)} - K_{i-1}^{(m)}) \rightarrow 0$, $K_1^{(m)} \leq A$ and $K_m^{(m)} \geq B$. Complete the tails as in Section 4.1 and place masses at midpoints as in Section 4.2. Let $h_m > 0$. If $h_m \rightarrow 0$ and $\delta_m/h_m^{3/2} \rightarrow 0$, then both the midpoint estimator (17) and its twiced version (20) converge to $f_{\mathbb{Q}}$ uniformly on every compact subset of (A, B) . If the grid is equally spaced, the Gaussian Euler–Maclaurin argument of the proof yields the same conclusion under the weaker condition $\delta_m/h_m \rightarrow 0$.*

Proof. The midpoint estimator is $\hat{f}_{\mathbb{Q}} = -F(\nu_N * \kappa_h)'$, and $f_{\mathbb{Q}} = -FP''$. Split

$$(\nu_N * \kappa_h)' - P'' = ((\nu_N - P) * \kappa_h') + (P' * \kappa_h - P')'.$$

The second term is ordinary kernel bias of $P'' = -f_{\mathbb{Q}}/F$: $(P' * \kappa_h - P')' = (P'' * \kappa_h) - P''$. Because κ is even, $\int \kappa = 1$ and $\int u\kappa(u) du = 0$. If $f_{\mathbb{Q}}''$ is bounded on a neighborhood of $[A, B]$, the difference is $O(h^2)$ uniformly on compact subsets of (A, B) several bandwidths away from the endpoints. The twiced kernel $2\kappa_h - \kappa_{h\sqrt{2}}$ has Fourier transform (19), hence vanishing second moment, so the same expansion starts at order h^4 whenever $f_{\mathbb{Q}}$ is C^4 .

The first term is discretization. On each completed cell the midpoint rule integrates $\kappa_h'(K - \cdot)$ against dP . For a C^2 integrand the error on a cell of length at most δ is $O(\delta^3)$ times the second derivative of the integrand. That second derivative is $O(h^{-4})$ for κ_h' , and $O(h^{-4})$ as well for the twiced equivalent kernel. The kernel window contains $O(h/\delta)$ cells, so a polynomial bound on the total discretization error is $O(\delta^2/h^3)$, which vanishes if $\delta/h^{3/2} \rightarrow 0$. For a Gaussian integrand on an equally spaced grid the Euler–Maclaurin formula (the Fourier transform of a Gaussian is a Gaussian) makes the midpoint error smaller than any power of δ/h , so the weaker condition $\delta/h \rightarrow 0$ already suffices. The càdlàg placement of Section 3.1, with masses at the right endpoints, is a rectangle rule. The error on a cell is $O(\delta^2)$ times the first derivative of the integrand, and that derivative is $O(h^{-3})$. Over the $O(h/\delta)$ cells in the kernel window the total is $O(\delta/h^2)$, which vanishes if $\delta/h^2 \rightarrow 0$.

The filled wings are supported on $(0, K_1]$ and, if $C_m > 0$, on $(K_m, K_{\text{end}}]$. Their contribution at an evaluation point $K \in [A + \varepsilon, B - \varepsilon]$ is at most a constant times $h^{-2} \exp(-\varepsilon^2/2h^2)$, because $K_1 \leq A$ and $K_m \geq B$. That bound tends to zero faster than any power of h .

Multiplying by $-F$ yields the claim for $\hat{f}_{\mathbb{Q}}$ and for $\hat{f}_{\mathbb{Q}}^{\text{tw}}$. $\square$

Remark 1. *The argument treats quotes as exact. It is not a statement about observation error, and it is not a statement about the far tails outside $[A, B]$. The equivalent kernel of (20) takes negative values, so the twiced estimator is not necessarily a density. Equation (21) is the complementary global statement: over the whole line the signed estimator integrates to zero.*

6.1. Numerical Illustration

The data-generating process is the stochastic-volatility model of Heston (1993) with the equity parameters of Bakshi et al. (1997): $S_0 = 100$, $r = 0.05$, $q = 0$, $T = 1/4$, $\kappa = 1.15$, $\theta = 0.04$, $\sigma = 0.39$, $\rho = -0.64$, $v_0 = 0.04$. The c.f. of $\log S_T$ uses the little-trap branch of Albrecher et al. (2007). Call prices are the

damped integrand of Carr and Madan (1999) with damping $\alpha = 1.5$, evaluated by the trapezoid rule on 4096 frequencies in $(0, 250]$; puts follow by parity. There is no Monte Carlo noise. The true density is Fourier inversion of the same c.f. on the same frequency grid,

$$f_{\mathbb{Q}}(K) = \frac{1}{K\pi} \int_0^\infty \operatorname{Re}[e^{-iu \log K} \varphi(u)] du,$$

where $\varphi(u) = \mathbb{E}^{\mathbb{Q}}[e^{iu \log S_T}]$. Two chains are used: a dense grid of $m = 256$ equally spaced strikes on $[30, 220]$ (spacing $\delta = 0.745$) and a sparse grid of $m = 64$ equally spaced strikes on $[70, 140]$ (spacing $\delta = 1.11$). Integrated squared error (ISE) is $\int_{60}^{150} (\hat{f}_{\mathbb{Q}} - f_{\mathbb{Q}})^2 dK$, evaluated by the trapezoid rule on 401 equally spaced points. Table 1 reports ISE for the four nested estimators at the oracle h^* , at the mesh rule $h = 1.2\delta$ of (22), and at the density-scale rule (23). The oracle minimizes the integral on a geometric grid in h .

Table 1: Integrated squared error $\int_{60}^{150} (\hat{f}_{\mathbb{Q}} - f_{\mathbb{Q}})^2 dK$ on exact Heston quotes. Mesh h is (22). The last pair of columns is (23). Baseline is (11) on the quoted strikes. Tail completion uses the wings of Section 4.1. Tail+Midpoint is (17). Tail+Mid+Twice is (20).

Chain	Estimator	h^*	Oracle ISE	Mesh h	Mesh ISE	Den. h	Den. ISE
Dense	Baseline (11)	0.63	2.67×10^{-5}	0.89	2.71×10^{-5}	3.41	1.45×10^{-4}
Dense	Tail completion	0.63	2.67×10^{-5}	0.89	2.71×10^{-5}	3.41	1.45×10^{-4}
Dense	Tail+Midpoint (17)	0.63	2.27×10^{-7}	0.89	8.23×10^{-7}	3.41	1.22×10^{-4}
Dense	Tail+Mid+Twice (20)	0.72	3.44×10^{-9}	0.89	5.00×10^{-9}	3.41	8.24×10^{-6}
Sparse	Baseline (11)	26.83	1.07×10^{-2}	1.33	2.87×10^2	4.51	7.42
Sparse	Tail completion	1.10	6.24×10^{-5}	1.33	6.34×10^{-5}	4.29	3.16×10^{-4}
Sparse	Tail+Midpoint (17)	0.96	4.39×10^{-6}	1.33	5.95×10^{-6}	4.29	2.68×10^{-4}
Sparse	Tail+Mid+Twice (20)	2.14	2.07×10^{-6}	1.33	3.24×10^{-6}	4.29	3.09×10^{-5}

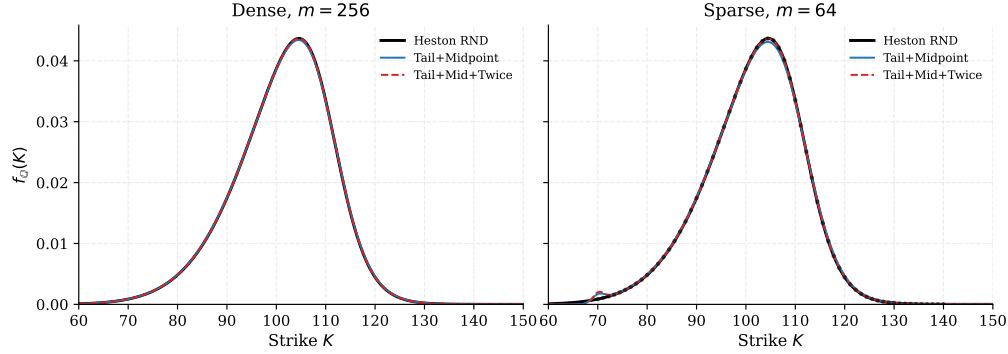


Figure 1: Heston density of S_T on $[60, 150]$. Solid: Tail+Midpoint (17) at mesh $h = 1.2\delta$. Dashed: Tail+Mid+Twice (20). Sparse quoted strikes are marked on the true density. Curves are truncated at zero.

On the dense chain $K_1 = 30$ lies well to the left of the ISE window and $C_m = 0$; completing the tails does not change the baseline digits, and the mesh rule already matches that oracle (ratio 1.02). Midpoints cut mesh ISE by a factor of thirty, to 8.23×10^{-7} against an oracle 2.27×10^{-7} (ratio 3.6). Twicing then reaches mesh ISE 5.00×10^{-9} against an oracle 3.44×10^{-9} (ratio 1.45), and is visually indistinguishable from the true density in Figure 1. On the sparse chain $K_1 = 70$ lies inside the window and $P(K_1) = 0.69$. Placing that mass at K_1 makes the mesh rule unusable (ISE 287). The left line (16) with spacing equal to the quoted mesh brings càdlàg mesh ISE to 6.34×10^{-5} . Midpoints on the completed support are Tail+Midpoint: mesh ISE 5.95×10^{-6} against oracle 4.39×10^{-6} (ratio 1.36). Twicing at the mesh h then reaches 3.24×10^{-6} against an oracle 2.07×10^{-6} (ratio 1.56).

The density-scale rule (23) is larger than the mesh on both chains ($h = 3.41$ on the dense grid, $h = 4.29$ on the sparse grid after the right wing). On exact dense quotes it oversmooths: twiced ISE 8.24×10^{-6} against mesh 5.00×10^{-9} . On the sparse chain it is larger than both the twiced oracle and twicing at the mesh (3.09×10^{-5} against 2.07×10^{-6} and 3.24×10^{-6}). It is the rule used on listed chains in Section 8, where the quotes are noisy.

Table 2 repeats the design for the variance-gamma process of Madan et al. (1998). The process parameters are those of Carr and Madan (1999): $\sigma = 0.12$, $\nu = 0.2$, $\theta = -0.14$. Spot, rate, dividend yield, and maturity are as above, as are the two strike grids and the ISE window. Variance gamma is a pure-jump Lévy process. The true density of $\log S_T$ is a modified Bessel function (Madan et al., 1998); the density of S_T is that formula times $1/K$. Calls are the discounted payoff

of that density, evaluated by the trapezoid rule. There is no Monte Carlo noise.

Table 2: Integrated squared error $\int_{60}^{150} (\hat{f}_Q - f_Q)^2 dK$ on exact variance-gamma quotes. Bandwidth columns as in Table 1.

Chain	Estimator	h^*	Oracle ISE	Mesh h	Mesh ISE	Den. h	Den. ISE
Dense	Baseline (11)	0.63	2.14×10^{-4}	0.89	2.42×10^{-4}	2.18	8.57×10^{-4}
Dense	Tail completion	0.63	2.14×10^{-4}	0.89	2.42×10^{-4}	2.18	8.57×10^{-4}
Dense	Tail+Midpoint (17)	0.54	1.69×10^{-5}	0.89	5.55×10^{-5}	2.18	7.18×10^{-4}
Dense	Tail+Mid+Twice (20)	0.63	2.88×10^{-6}	0.89	9.96×10^{-6}	2.18	2.18×10^{-4}
Sparse	Baseline (11)	26.83	2.80×10^{-2}	1.33	2.87×10^2	2.88	2.86×10^1
Sparse	Tail completion	0.84	4.67×10^{-4}	1.33	5.66×10^{-4}	2.70	1.53×10^{-3}
Sparse	Tail+Midpoint (17)	0.84	5.23×10^{-5}	1.33	1.91×10^{-4}	2.70	1.26×10^{-3}
Sparse	Tail+Mid+Twice (20)	0.84	1.17×10^{-5}	1.33	4.45×10^{-5}	2.70	4.44×10^{-4}

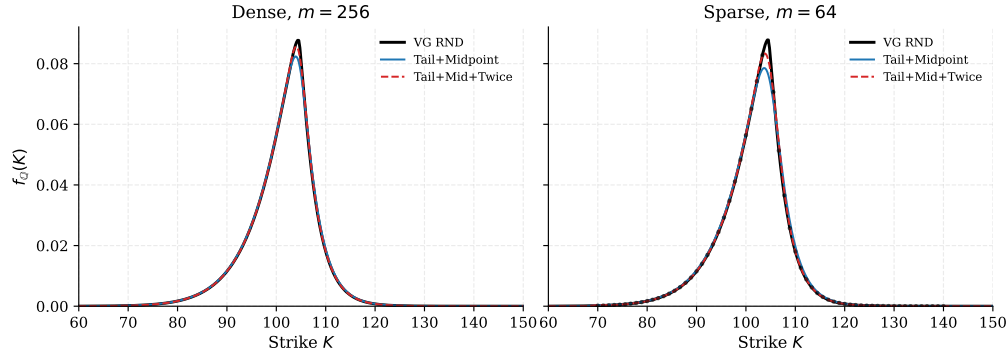


Figure 2: Variance-gamma density of S_T on $[60, 150]$. Solid: Tail+Midpoint (17) at mesh $h = 1.2\delta$. Dashed: Tail+Mid+Twice (20). Sparse quoted strikes are marked on the true density. Curves are truncated at zero.

On the dense chain the ranking is the same as under Heston. Completing the tails does not change the baseline digits. Midpoints cut mesh ISE from 2.42×10^{-4} to 5.55×10^{-5} . Twicing then reaches 9.96×10^{-6} against an oracle 2.88×10^{-6} (ratio 3.5), and tracks the peak in Figure 2. The density of S_T is more peaked than under Heston, and the absolute errors are larger. On the sparse chain the càdlàg mesh rule is again unusable (ISE 287). Tail+Midpoint mesh ISE is 1.91×10^{-4} ; twicing at that h lowers it to 4.45×10^{-5} against an oracle 1.17×10^{-5} . The density-scale rule oversmooths on both chains.

7. Observation Noise

The chains in Section 6 are exact and equally spaced. A listed chain is neither. This section keeps the Heston and variance-gamma targets and changes the design in three ways. Strikes are irregular: twenty-two geometrically spaced knots on $[0.45F, 0.90F)$, twenty-eight equally spaced knots on $[0.90F, 1.10F)$, and twenty-two geometrically spaced knots on $[1.10F, 1.40F]$, seventy-two strikes in all. The first strike lies inside the integrated-squared-error window $[60, 140]$. Black–Scholes implied volatility is observed with independent Gaussian noise of standard deviation 0.01, then floored at 0.02, and converted back to a call. Thirty replications use the seed 20260923. No method sees $f_{\mathbb{Q}}$ when it chooses a tuning parameter.

The estimator of this paper is scored three times. On the raw quotes, the density-scale rule (23) is the rule used on listed chains, and a second bandwidth is chosen by even/odd cross-validation of out-of-the-money pricing error. The two-step version builds a different call curve and then uses the same kernel at the density-scale bandwidth. The quoted calls are replaced by their Euclidean projection onto the decreasing convex cone with slopes in $[-e^{-rT}, 0]$, the quadratic program of Section 8. Beyond the first and last strikes, implied volatility is held at its wing value and the calls are repriced on a log grid from $0.15F$ to $3F$. The mixture uses those same masses. It puts weight α on the twiced Gaussian at bandwidth ch and weight $1 - \alpha$ on a sinc kernel at bandwidth $s \cdot ch$, where h is the density-scale bandwidth of the training fold. The triple (c, α, s) is chosen by even/odd cross-validation of out-of-the-money pricing error on the grid $c \in \{0.75, 1, 1.25, 1.5\}$, $\alpha \in \{\frac{1}{2}, \frac{3}{4}, 1\}$, and $s \in \{2, 3\}$ when $\alpha < 1$. The pilot row freezes $c = 1$, $\alpha = \frac{1}{2}$, and $s = 3$, the triple with the lowest integrated squared error on the first eight replications of each design. Out-of-the-money premiums are floored at zero before the error is computed. The other methods are:

- Yatchew and Härdle, with the roughness penalty chosen by the same cross-validation. The unknown in the quadratic program is the call. Slope increments are not used: that map is ill-conditioned, and a sequential least-squares solver started from those increments stops on an infeasible subproblem and returns the isotonic start. In call coordinates the same program terminates at the projection. The fitted call is convex and decreasing, with slopes in $[-e^{-rT}, 0]$, so its second derivative is a discrete measure. It is scored on pricing error and not on integrated squared error.

- Aït-Sahalia and Duarte: the same projection, then a local linear whose bandwidth is cross-validated, and a local cubic at $0.9 F \hat{\sigma} \sqrt{T} n^{-1/9}$ for the density.
- Bondarenko, forty-one lognormals on a uniform grid in log strike from the lowest quoted strike to the highest. The common width is cross-validated on a grid that starts at the center spacing. Weights are nonnegative. Sum-to-one and forward-mean conditions enter as penalties and the weights are not rescaled.
- Priestley and Chao, the cubic of Section 3.2, at two bandwidths: the cross-validated pricing bandwidth, and $1.06 F \hat{\sigma}_{\text{ATM}} \sqrt{T} n^{-1/9}$.
- A raw SVI slice (Gatheral, 2006). Total implied variance is $a + b(\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2})$, fitted to call prices by least squares. The butterfly function $g(k)$ of Gatheral and Jacquier (2014) is penalized wherever it is negative.

Integrated squared error is $\int_{60}^{140} (\hat{f}_{\mathbb{Q}} - f_{\mathbb{Q}})^2 dK$, on the signed estimator. Pricing error is out-of-the-money root-mean-square error against the clean calls, not against the noisy quotes.

Table 3: Noisy irregular grids, thirty replications. The first line of each method is the mean and the second is the replication standard deviation. ISE is $\int_{60}^{140} (\hat{f}_{\mathbb{Q}} - f_{\mathbb{Q}})^2 dK$. RMSE is out-of-the-money pricing error against the clean calls. Yatchew–Härdle returns a convex call, not a density curve.

Method	Rule	Heston		Variance gamma	
		ISE	RMSE	ISE	RMSE
Mixture	pricing CV	6.62×10^{-4}	3.21×10^{-2}	4.23×10^{-3}	3.49×10^{-2}
		8.76×10^{-4}	1.12×10^{-2}	3.57×10^{-3}	8.70×10^{-3}
Mixture	pilot, $\alpha = \frac{1}{2}, 3h$	2.79×10^{-4}	3.08×10^{-2}	1.27×10^{-3}	2.94×10^{-2}
		1.31×10^{-4}	8.20×10^{-3}	5.52×10^{-4}	7.80×10^{-3}
Two-step	projection, flat wing	4.29×10^{-4}	2.91×10^{-2}	2.35×10^{-3}	3.17×10^{-2}
		2.63×10^{-4}	9.10×10^{-3}	1.34×10^{-3}	8.00×10^{-3}
Kernel	density scale	5.72×10^{-3}	3.68×10^{-2}	8.94×10^{-2}	4.22×10^{-2}
		7.59×10^{-3}	1.36×10^{-2}	1.15×10^{-1}	1.40×10^{-2}
Kernel	pricing CV	7.72×10^{-2}	3.82×10^{-2}	1.49×10^{-1}	4.02×10^{-2}
		2.61×10^{-1}	1.51×10^{-2}	6.05×10^{-1}	1.09×10^{-2}
Yatchew–Härdle	CV penalty	—	4.22×10^{-2}	—	4.16×10^{-2}
		—	9.29×10^{-3}	—	9.25×10^{-3}
Aït-Sahalia–Duarte	CV / $n^{-1/9}$	4.67×10^{-4}	4.31×10^{-2}	2.01×10^{-3}	3.81×10^{-2}
		1.18×10^{-4}	1.19×10^{-2}	6.20×10^{-4}	1.08×10^{-2}
PCA	CV width	8.11×10^{-4}	3.29×10^{-2}	1.70×10^{-3}	3.76×10^{-2}
		1.34×10^{-3}	9.97×10^{-3}	9.92×10^{-4}	1.77×10^{-2}
Priestley–Chao	pricing CV	2.51×10^{-2}	4.50×10^{-2}	1.85×10^{-2}	4.16×10^{-2}
		1.50×10^{-2}	1.10×10^{-2}	1.28×10^{-2}	1.51×10^{-2}
Priestley–Chao	$n^{-1/9}$	8.92×10^{-4}	4.34×10^{-1}	3.14×10^{-3}	2.34×10^{-1}
		1.46×10^{-4}	3.15×10^{-2}	6.52×10^{-4}	2.95×10^{-2}
SVI	price fit	2.46×10^{-3}	2.86×10^{-2}	2.26×10^{-2}	2.87×10^{-2}
		4.95×10^{-3}	1.05×10^{-2}	2.61×10^{-2}	8.07×10^{-3}

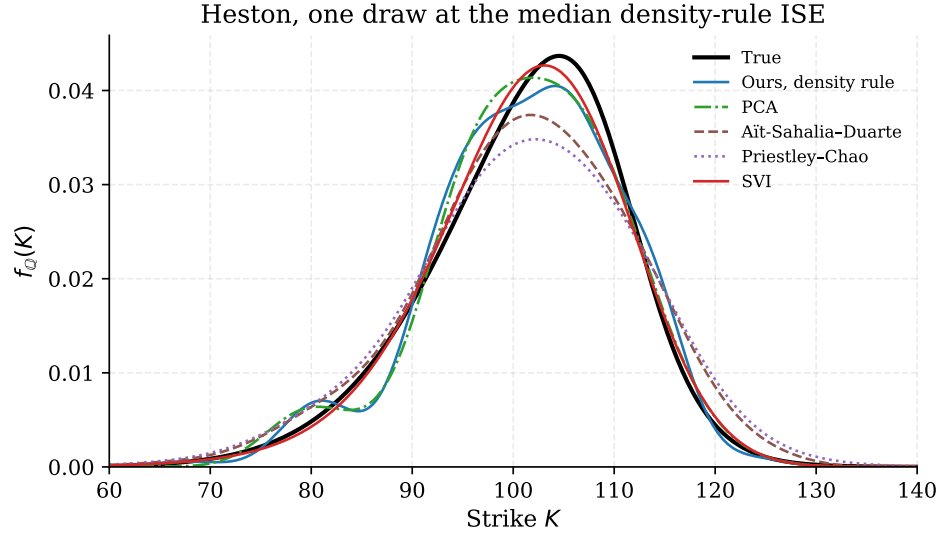


Figure 3: One Heston draw, the replication whose density-scale integrated squared error is the sample median. Curves other than the truth are truncated at zero for display. Priestley–Chao is shown at the second-derivative bandwidth.

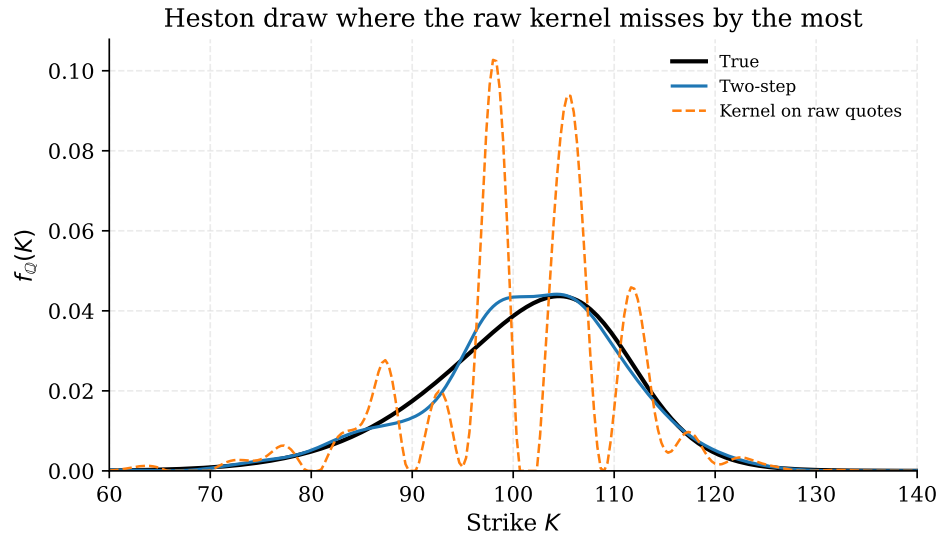


Figure 4: The Heston replication in which the raw kernel's integrated squared error exceeds the two-step estimator by the most: 0.025 against 2.9×10^{-4} . Curves are truncated at zero for display.

On the raw quotes the local cubic and the retuned convolution are closer to f_Q than the density-scale kernel. Cross-validating that kernel on pricing error makes the density worse: under Heston the mean integrated squared error rises from 5.7×10^{-3} to 7.7×10^{-2} . The second-derivative Priestley–Chao bandwidth has Heston pricing error 0.43; the same cubic at a pricing bandwidth has pricing error 0.045 and a worse density.

Projecting the calls and holding the wings at the wing implied volatility is the step that removes the arbitrage noise in the jumps. The twiced Gaussian on that curve has mean integrated squared error 4.3×10^{-4} under Heston, below the local polynomial (4.7×10^{-4}) and the convolution (8.1×10^{-4}), and 2.3×10^{-3} under variance gamma, above both. Figure 4 is the Heston draw on which the raw kernel is worst: integrated squared error 0.025, against 2.9×10^{-4} once the same kernel is applied to the projected call.

Moving half of that kernel onto a sinc at three times the bandwidth, with the constants chosen by integrated squared error on the first eight replications, lowers the thirty-replication means to 2.8×10^{-4} and 1.3×10^{-3} . That is the smallest integrated squared error in Table 3, on both targets, and the clean-call error is 0.031 and 0.029 against 0.029 for SVI. Those two constants see the true density on the pilot.

Pricing cross-validation is run only after the projection, the wing, and the mixture are in place. It chooses (c, α, s) from the grid in the paragraph above, using held-out quotes and not f_Q . The modal Heston choice is $c = 1.25$, $\alpha = \frac{1}{2}$, $s = 2$ (nine of thirty draws). The mean integrated squared error rises to 6.6×10^{-4} under Heston and 4.2×10^{-3} under variance gamma. The Heston median is 4.0×10^{-4} , so the mean is a few draws on which the cross-validation narrows the kernel. Under Heston the cross-validated mixture still beats the convolution and does not beat the local polynomial. Under variance gamma it beats neither. Clean-call error stays near 0.03. The projection’s second derivative remains a discrete measure. The kernel, Gaussian or mixed, is the step that returns $\hat{f}_Q$.

8. Empirical Illustration

8.1. Data

Four slices illustrate the estimator. They are not a panel. The S&P 500 snapshot is 6 September 2026, 17:04 Eastern; the Nasdaq-100 snapshot is 8 September 2026, 17:17 Eastern; the Russell 2000 snapshot is 19 September 2026, 03:44 Eastern, outside the cash session. Only the standard monthly roots are kept. Time to expiry is $T = d/365.25$. The discount rate is a flat $r = 0.04$, and the forward is the

median of $K + e^{rT}(C - P)$ near the money, so an error in r moves F . The quote filter is unchanged: a positive bid, ask at least the bid, mid at least 0.25, and spread at most $\max(15, 0.5 \times \text{mid})$. Out-of-the-money mids are kept and the other side is filled by parity. Puts with $K < 0.7F$ and mid below 0.50 are dropped. Table 4 records the counts.

The December 2026 S&P 500 strip ($T = 0.282$, $S_0 = 7718.60$, $F = 7786$, $q = 0.92\%$) has 393 strikes on $[0.31F, 1.36F]$ after the screens. The March 2027 strip ($T = 0.531$, $F = 7864$) has 222. The Nasdaq-100 December strip ($T = 0.277$, $F = 29898$) has 252. The Russell 2000 December strip ($T = 0.246$, $F = 2883$) has 97. Density-scale bandwidths are $h = 180, 291, 1082$, and 97.8.

Table 4: Construction of the working slices. “At expiry” counts listed puts and calls on that root and maturity with a positive bid and ask at least the bid. “Screens” applies the mid and spread filters. “Working OTM” is the slice used by the estimators.

Slice	At expiry	Screens	Unique K	Working OTM	Puts
SPX 18 Dec 2026	819	812	416	393	303
SPX 19 Mar 2027	453	452	228	222	164
NDX 18 Dec 2026	555	531	279	252	185
RUT 18 Dec 2026	217	216	115	97	56

8.2. Pricing and mass

Every method is tuned only on pricing error, by the nested even/odd rule of Section 7: on each half-sample the tuning parameter is chosen from that half alone, and the held-out strikes are scored once. Premiums are floored at zero. Table 5 reports in-sample and hold-out root-mean-square error, and the positive and negative parts of $\hat{f}_{\mathbb{Q}}$ on $[\max(1, 0.15F), 2.6F]$. The Yatchew–Härdle row has no density column. The convex call’s second derivative is atomic.

Table 5: Listed slices. RMSE and hold-out are out-of-the-money root-mean-square errors, in index points. Hold-out retunes inside each training half. Positive and negative masses are integrals of $\max(\hat{f}_Q, 0)$ and $\min(\hat{f}_Q, 0)$ on $[\max(1, 0.15F), 2.6F]$.

Slice	Method	RMSE	Hold-out	Positive	Negative
SPX Dec	Kernel, density scale	4.88×10^{-1}	9.46×10^{-1}	1.01	-1.40×10^{-2}
	Two-step	4.99×10^{-1}	8.69×10^{-1}	1.01	-1.20×10^{-2}
	Mixture, pricing CV	3.29×10^{-1}	5.71×10^{-1}	1.12	-1.10×10^{-1}
	Kernel, pricing CV	4.15×10^{-1}	2.86×10^{-1}	4.28×10^1	-4.18×10^1
	Yatchew–Härdle	2.20×10^{-2}	5.40×10^{-2}	—	—
	Aït-Sahalia–Duarte	4.11×10^{-1}	5.40×10^{-1}	1.03	-3.85×10^{-1}
	PCA	3.96×10^{-2}	4.28×10^{-2}	1.00	0.00
	Priestley–Chao, pricing CV	9.08×10^{-2}	3.48×10^{-1}	1.01	-1.21×10^{-2}
	Priestley–Chao, $n^{-1/9}$	2.05×10^1	2.37×10^1	9.99×10^{-1}	-1.07×10^{-2}
	SVI	5.32×10^{-1}	5.32×10^{-1}	1.00	0.00
SPX Mar	Kernel, density scale	9.55×10^{-1}	1.90	1.02	-2.11×10^{-2}
	Two-step	9.40×10^{-1}	1.72	1.01	-2.20×10^{-2}
	Mixture, pricing CV	5.12×10^{-1}	1.32	1.07	-7.40×10^{-2}
	Kernel, pricing CV	7.36×10^{-1}	5.24×10^{-1}	4.02×10^1	-3.92×10^1
	Yatchew–Härdle	3.80×10^{-2}	1.41×10^{-1}	—	—
	Aït-Sahalia–Duarte	6.51×10^{-1}	8.61×10^{-1}	1.02	-3.21×10^{-7}
	PCA	8.73×10^{-2}	1.02×10^{-1}	1.00	0.00
	Priestley–Chao, pricing CV	3.22×10^{-1}	1.28	1.00	-2.14×10^{-3}
	Priestley–Chao, $n^{-1/9}$	2.86×10^1	3.30×10^1	9.97×10^{-1}	-3.51×10^{-1}
	SVI	1.16	1.16	1.00	-1.71×10^{-8}
NDX Dec	Kernel, density scale	6.45	8.74	1.16	-1.61×10^{-1}
	Two-step	6.42	8.82	1.16	-1.63×10^{-1}
	Mixture, pricing CV	5.93	7.83	1.29	-2.99×10^{-1}
	Kernel, pricing CV	6.66	6.86	1.14	-1.44×10^{-1}
	Yatchew–Härdle	2.06	3.05	—	—
	Aït-Sahalia–Duarte	3.72	4.73	1.03	-2.10×10^{-2}
	PCA	2.93	3.23	1.00	0.00
	Priestley–Chao, pricing CV	2.88	7.41	1.02	-2.44×10^{-2}
	Priestley–Chao, $n^{-1/9}$	1.36×10^2	1.57×10^2	9.99×10^{-1}	-3.28×10^{-1}
	SVI	3.12	3.29	1.00	0.00
RUT Dec	Kernel, density scale	4.03×10^{-1}	9.64×10^{-1}	1.01	-7.04×10^{-3}
	Two-step	4.46×10^{-1}	8.14×10^{-1}	1.00	-2.00×10^{-3}
	Mixture, pricing CV	3.58×10^{-1}	6.27×10^{-1}	1.10	-1.21×10^{-1}
	Kernel, pricing CV	5.30×10^{-1}	7.44×10^{-1}	1.01×10^1	-9.06
	Yatchew–Härdle	2.30×10^{-2}	1.55×10^{-1}	—	—
	Aït-Sahalia–Duarte	3.60×10^{-1}	4.94×10^{-1}	9.95×10^{-1}	-1.98×10^{-1}
	PCA	2.92×10^{-2}	3.29×10^{-2}	1.00	0.00
	Priestley–Chao, pricing CV	2.11×10^{-1}	8.36×10^{-1}	9.99×10^{-1}	-9.74×10^{-4}
	Priestley–Chao, $n^{-1/9}$	1.38×10^1	1.59×10^1	9.98×10^{-1}	-2.49×10^{-1}
	SVI	1.17×10^{-1}	1.17×10^{-1}	1.00	0.00

On these four slices the Gaussian two-step tracks the raw density-scale kernel. The quotes are already close to the cone, so the projection moves them by hundredths of an index point and the density-scale bandwidth smooths either curve the same way. December S&P 500 error is 0.50 against 0.49. Pricing cross-validation of the mixture, the same grid as in Section 7, lowers that strip to 0.33 in sample and 0.57 on the hold-out. The positive mass on the window is 1.12 and the negative part is -0.11 . The unsmoothed projection is still the closest call, at 0.022, and the convolution is at 0.040. The same ordering holds on the other three slices: cross-validation improves the mixture relative to the fixed wide sinc, and the projection and the convolution remain closer to the quotes.

The convex projection is the closest of these methods to the quotes. On the December S&P 500 strip its in-sample error is 0.022 and its hold-out error is 0.054. The cross-validated penalty is 1 in the normalized units of Section 7, not zero: a pure interpolating projection is available and is not what the hold-out selects. The same pattern holds on the other three slices (penalties 1, 0.1, and 10). The quotes are close to the decreasing convex cone once the quadratic program is solved in the call. An earlier implementation, parameterized by slope increments, stopped with an infeasible subproblem and returned an isotonic start whose December S&P 500 residual was about 2.25. That number is not the distance to the cone.

The positive convolution, with centers on the quoted log-strike range and a width at the bottom of the cross-validation grid, is the next pricing method on every slice, and on the Russell 2000 strip it matches the projection on hold-out (0.033 against 0.155). It integrates to one, up to the penalty residual. SVI does as well. Its butterfly function is nonnegative on the monitoring grid, up to a solver tolerance of 3×10^{-7} on the March S&P 500 strip. Both are smoother than the quotes and both are probabilities.

The density-scale rule reprices the December S&P 500 strip with error 0.49, and the Nasdaq-100 strip with error 6.45. Both numbers sit above the convex projection and above the convolution. On $[\max(1, 0.15F), 2.6F]$ the signed mass is 1.00 and the positive mass is 1.01 on the S&P 500 and Russell 2000 strips. On the Nasdaq-100 strip the positive mass is 1.16 and the negative mass inside the same window is -0.16 . On a grid $[-F, 8F]$ the signed integral is 0 and the first moment equals F , to a forward error below 10^{-3} index points on each of the four slices. The positive and negative parts on that wide grid each integrate to between 1.07 and 1.23. Equation (21) is not an asymptotic remark. It is the estimator.

Pricing cross-validation does not repair the mass. On the December S&P 500 strip it selects $h = 20$, against a density-scale bandwidth of 180. In-sample error

falls from 0.49 to 0.42 and hold-out error from 0.95 to 0.29, and the positive mass on the window rises to 42.8. The March strip and the Russell 2000 strip do the same thing, with positive masses 40.2 and 10.1. The Nasdaq-100 cross-validation stays near the density-scale bandwidth ($h = 1126$ against 1082) and the mass stays near 1.14. A bandwidth chosen for the call is not a bandwidth for the density. Figure 5 plots the density-scale estimator, not the cross-validated one.

The Priestley–Chao row at $n^{-1/9}$ reproduces the large pricing errors of an untuned second-derivative bandwidth: 20.48 on the December S&P 500 strip and 135.69 on the Nasdaq-100 strip. At the pricing bandwidth those errors are 0.09 and 2.88. The local polynomial of Aït-Sahalia and Duarte, started from the solved projection rather than from an isotonic start, has December S&P 500 error 0.41. Its hold-out error is larger than its in-sample error on every slice. The local cubic is a poor probability on the extended window: the December S&P 500 negative mass is -0.39 , and the first moment is not the forward. That moment is not reported in the table because the cubic is being integrated outside the region where a local polynomial of a convex call is a density.

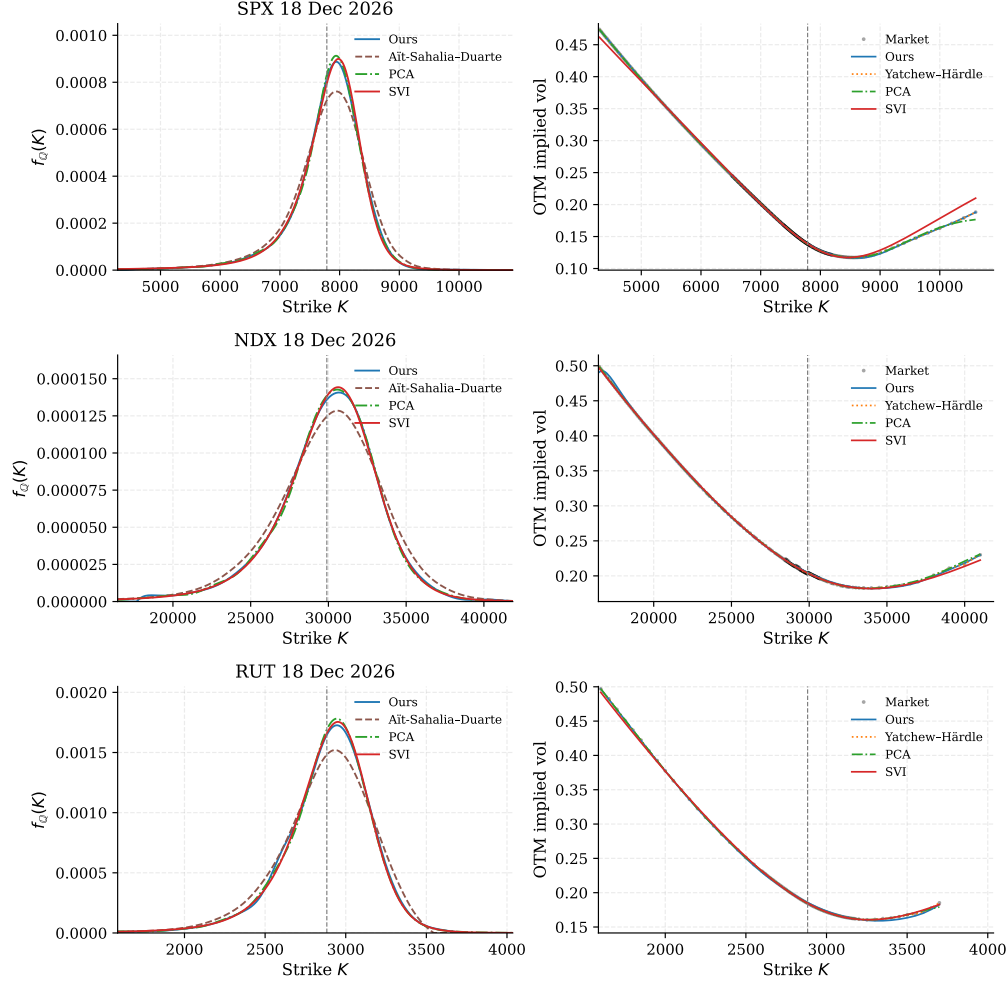


Figure 5: December 2026 slices. Left: positive part of the density-scale estimator, the local cubic, the convolution, and SVI. Right: out-of-the-money implied volatility. The dashed line is the forward.

9. Conclusion

The second strike derivative of a European call is the risk-neutral density of the underlying at expiry. On a finite strike list that derivative is not an estimator. Rescaling the calls to a complementary distribution function makes the object to be smoothed a jump measure. A Gaussian kernel then has a closed form: the density estimator is minus the forward times the derivative of a Gaussian mixture of those jumps, and the call interpolant is the mixture's cdf. Completing the tails,

placing masses at cell centers, and combining two widths are quadrature corrections. On exact equally spaced Heston quotes they bring the integrated squared error to 5.0×10^{-9} . The same ordering holds for variance gamma.

Over the whole line the signed estimator integrates to zero and its first moment equals the forward, before and after twicing. Uniform consistency on a compact interior interval does not put the missing unit of mass back. Prices taken from the clipped cdf are not integrals of the density.

On exact quotes the quadrature corrections bring the integrated squared error to 5.0×10^{-9} . On noisy quotes the jumps of the raw call are the wrong measure. Projecting the calls and holding the wings at the wing implied volatility, then applying the twiced Gaussian, lowers the Heston mean to 4.3×10^{-4} . An equal Gaussian–sinc mixture with the sinc at three times the bandwidth, with those constants chosen by integrated squared error on eight pilot replications, lowers the thirty-replication means further, to 2.8×10^{-4} and 1.3×10^{-3} . Pricing cross-validation of the same three constants, after the projection and the wing are already in place, does not recover that pair. The means rise to 6.6×10^{-4} and 4.2×10^{-3} . Under Heston the cross-validated mixture still beats the convolution. Under variance gamma it does not beat the convolution or the local polynomial. On listed slices the quotes already lie near the cone. Cross-validation lowers the December S&P 500 mixture error from the fixed wide sinc to 0.33, and the unsmoothed projection remains at 0.022.

The estimator is a fast local approximation of f_Q on a fine exact strip. Under noise, the lowest integrated squared error in the comparison uses the projected call and a Gaussian–sinc mixture whose constants were chosen for that error. Choosing the constants from prices instead gives back part of the gain. Neither version is the closest interpolant of a listed chain that is already nearly convex.

Code and Data Availability

A Python implementation of (20) and the replication scripts are available at <https://github.com/s-broda/RND>. The v2 comparison is `python/replicate_v2.py`. The two-step mixture is `python/src/twostep.py`.

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